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REPORT

Title: Naming Fukushima: A Critical Discourse Analysis of Media Labels for Japan's Wastewater Discharge.

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I. Background

1.1 Brief history of Fukushima discharge

The Fukushima Dai'ichi Nuclear Power Station (FDNPS) suffered core meltdowns in three reactors after the Tōhoku earthquake and tsunami on 11 March 2011. The accident produced large amounts of contaminated cooling water, and continued groundwater inflow has added to the volume stored in on-site tanks (Mabon, 2024, p. 1). The plant operator, Tokyo Electric Power Company (TEPCO, has processed much of this water through the Advanced Liquid Processing System (ALPS), which removes most radioactive elements. Tritium, a radioactive form of hydrogen that is not easily extracted, remains in the treated water (Mabon, 2024, p. 2; Akiko, 2024).

With storage space running out and more than a million tonnes of water accumulated, the Japanese government authorized a plan in 2021 to gradually discharge the ALPS-treated and diluted water into the Pacific Ocean over the coming decades. The first controlled releases started in August 2023 (METI, 2023; Mabon, 2024, p. 2). After a comprehensive review, the International Atomic Energy Agency (IAEA) concluded that the planned discharge, if executed and monitored as designed, will meet international safety standards and have a negligible radiological impact. (IAEA, 2023; METI, 2023). The IAEA has since continued to report that the process conforms to international guidelines, underscoring the importance of ongoing monitoring and transparency (AP, 2024).

1.2 Controversy over terminology

The planned discharge of water from the Fukushima Daiichi plant has been accompanied by significant international debate. A central and persistent feature of this debate is the divergence in the terminology used by different national actors and media systems to describe the water itself. These labeling differences are not incidental but form a core part of the public and diplomatic discourse surrounding the event.

1.2.1 Japan: “treated water” framing

Japan's official communications and mainstream media consistently employ the terms “treated water” (処理水) or “ALPS-treated water” (ALPS 処理水) (毎日新聞 [Mainichi Shimbun], 10/09/2023; 読売新聞 [Yomiuri Shimbun], 01/10/2023). This formulation presents the discharge as a managed, technical operation, emphasizing the water's processing through the ALPS. Official narratives highlight validation from the IAEA and reference international safety standards, aiming to assure the public and the international community of the action's safety and regulatory compliance. This terminology forms the cornerstone of Japan's public-facing discourse on the issue.

1.2.2 China: “contaminated/ radioactive water” framing

In stark contrast, Chinese official statements and state media consistently employ labels such as “核污染水” (nuclear-contaminated water), “核废水” (nuclear-wasted water) or “放射性污染” (radioactive pollution) (人民网 [People's Daily Online], 13/04/2022; 新华网 [Xinhua net], 24/08/2023; 央广网 [China National Radio], 23/08/2023). This terminology explicitly frames the discharge as a dual threat—to both the environment and regional geopolitical stability (Chen & Pan, 2025, p. 137).

1.2.3 Western Media: “technical / scientific naming

Coverage in major Western, English-language media outlets typically employs a range of terms including “treated water,” “radioactive water,” and “wastewater” (PBS NewsHour, 23/08/2023; Reuters, 22/08/2023; TIME, 06/02/2023). Reports often incorporate technical descriptors and reference regulatory bodies such as the IAEA. Headlines and articles frequently include details about the scale of the discharge, the treatment process, and the involvement of scientific institutions. The language in this coverage generally emphasizes procedural and regulatory aspects of the event.

1.3 Problem Statement

The stark divergence in terminology used to describe the Fukushima wastewater—from "treated water" to "nuclear-contaminated water"—constitutes a significant real-world problem with implications for international relations, public risk perception, and environmental governance. While scholars have documented differing national frames (e.g., Li & Lin, 2025; Chen & Liu, 2024), a scholarly gap persists. Existing research often catalogs labeling preferences or analyzes media narratives within single national contexts but lacks a systematic, comparative Critical Discourse Analysis (CDA) that simultaneously examines lexicalization, intertextuality, and actor representation across media systems.

Specifically, there is limited comparative research that simultaneously examines the interplay of three core CDA concepts: (1) Lexicalization (the strategic choice of labels), (2) Intertextuality (how media cite authoritative sources like the IAEA to construct credibility), and (3) Actor Representation (the portrayal of key actors to assign responsibility or blame). This gap leaves an incomplete understanding of the discursive mechanisms through which media in Japan, China, and Western nations produce such radically different narratives about the same scientific-environmental event.

Therefore, this study will conduct a targeted, comparative CDA of news media from these three regions to address this gap. It aims to provide a systematic linguistic explanation of how naming practices, source mobilization, and actor portrayals converge within each media system to construct distinct narratives of risk, scientific authority, and geopolitical stance. This approach moves beyond noting that frames differ to explain how they are produced through specific, interrelated discursive strategies.

II. Theory — Critical Discourse Analysis

This study adopts CDA as its theoretical and analytical framework to examine how Japanese, Chinese, and Western media construct meaning through their naming and framing of the Fukushima wastewater discharge. CDA views discourse as inherently social and ideological, highlighting how language shapes, and is shaped by, relations of power, institutional interests, and socio-political contexts. Rather than treating news discourse as neutral representation, CDA emphasizes that linguistic choices systematically privilege certain interpretations while marginalizing others (Fairclough, 1992; Van Dijk, 1993).

CDA is especially relevant for issues involving scientific uncertainty, environmental governance, and geopolitical tension. The Fukushima discharge embodies all three. What various actors choose to call the water: "treated water," "nuclear-contaminated water," "radioactive wastewater," is not simply descriptive terminology. These lexical choices perform ideological work: they frame the level of risk, assign or deflect blame, and position national or institutional actors within competing narratives of responsibility, safety, and harm.

Grounded in Fairclough's three-dimensional model, this study focuses primarily on textual analysis, examining how vocabulary, modality, tone, and representations of actors encode underlying assumptions and value systems. Three CDA concepts are central to the analytical process:

2.1 Lexicalization

Lexicalization refers to the deliberate choice of words that construct particular meanings and evaluative stances. In this context, competing labels such as "treated water" versus "nuclear-contaminated water" reveal contrasting ideological orientations: the former normalizes the discharge as scientifically managed, while the latter foregrounds danger and contamination. Such naming practices are central to shaping public perception and policy legitimacy.

2.2 Intertextuality

Intertextuality concerns how media texts draw upon, repeat, or transform the language of other sources—such as government agencies, scientific institutions, or international bodies like the IAEA. The authoritative voices that media choose to foreground or background play a key role in legitimizing or contesting particular interpretations of the wastewater discharge. For instance, Japanese news frequently cite TEPCO or the IAEA to reinforce credibility, whereas Chinese media rely heavily on domestic experts to validate risk-oriented narratives.

2.3 Actor Representation

CDA examines how social actors are constructed through inclusion or exclusion, attribution of agency, use of evaluative labels, and positioning within broader narratives. Media can depict actors such as the Japanese government, TEPCO, the IAEA, or fishermen as responsible, negligent, authoritative, vulnerable, or politically motivated. These representations are not neutral; they embed national interests, cultural values, and political agendas, shaping how audiences assign responsibility for environmental risk.

Collectively, these dimensions enable a nuanced analysis of how media in different sociopolitical contexts construct divergent narratives around the same environmental event. By scrutinizing naming practices, tone, imagery, and modality markers, CDA reveals how discourse contributes to shaping public meaning, amplifying or attenuating perceived risks, and reinforcing geopolitical alignments. In this study, CDA provides the conceptual tools needed to uncover how linguistic selections across Japanese, Chinese, and Western media articulate contrasting ideological positions amid scientific ambiguity and transnational responsibility.

III. Method

This study employs a qualitative CDA to examine how Japanese, Chinese, and Western media construct divergent narratives about Japan's release of treated wastewater from the Fukushima Daiichi Nuclear Power Station. CDA is well suited to uncover the ideological functions of language, enabling systematic exploration of how lexical choices, evaluative stances, and representational strategies naturalize particular interpretations of environmental risk and political responsibility.

3.1 Data Collection

A corpus of 33 news articles was assembled from three media regions:

- Japan: 11 articles from major national outlets such as NHK, Asahi Shimbun, Yomiuri Shimbun, and Mainichi Shimbun.
- China: 10 articles from state or semi-official sources including Xinhua, People's Daily, China Daily, and CNR.
- Western media: 12 articles from leading English-language platforms such as Reuters, BBC, AP News, Guardian, Scientific American, Financial Times, and Al Jazeera.

Articles were selected based on their relevance to the Fukushima discharge, explicit reference to naming or framing of the wastewater, and publication during the key decision-making and release period. This approach ensured a balanced and comparable dataset for cross-regional analysis.

3.2 Analytical Procedure

The analysis unfolded in two sequential phases:

Phase 1: Individual Article Coding

Each article was subjected to close textual reading to identify linguistic features that generate ideological meaning or portray social actors. The coding process emphasized micro-level textual elements (lexical choices, tone, modality markers, metaphors, and actor

descriptions) that contribute to discursive constructions of risk, responsibility, and legitimacy.

Phase 2: Cross-Regional Comparative Analysis

After coding individual texts, patterns were compared across Japanese, Chinese, and Western outlets. This stage focused on identifying how naming practices and textual strategies converged or diverged across regions, and how these differences reflected broader political, cultural, or institutional contexts.

3.3 Coding Framework

To enable systematic examination, a coding table (see Appendix B) was developed to catalog key textual features across news all articles. Analysis focused on five predefined categories:

1. Naming: The specific terms used for the water (e.g., "treated water," "radioactive water," "nuclear-contaminated water").
2. Tone: The evaluative stance towards the discharge, spanning from reassuring to alarmist.
3. Actor Representation: The portrayal of central entities, such as the Japanese government, TEPCO, the IAEA, and neighboring countries.
4. Metaphors and Imagery: Figurative language or vivid descriptions that amplify or downplay the sense of risk.
5. Modality: Expressions of certainty, probability, or obligation, using markers like "will," "may," "could," or "is expected to."

These categories are grounded in CDA textual analysis literature, which identifies lexicalization, semantic preference, actor representation, and modality as core signals of ideological positioning (Fairclough, 1992; Van Dijk, 1998). Coding the articles against these categories allowed for the identification of prevailing linguistic frames and discursive emphases that differentiate the media narratives from each region.

3.4 Synthesis and Interpretation

The final stage involved integrating coded findings to interpret the overarching discursive patterns characterizing each media region. This synthesis focused on how linguistic choices

articulated contrasting narratives about safety, legitimacy, scientific authority, and geopolitical positioning. The comparative CDA enabled the study to reveal how different media systems construct divergent meanings around the same environmental event.

IV. Findings and Discussions

This section presents a comparative critical discourse analysis of the media coverage in Japanese, Chinese, and Western outlets regarding the Fukushima wastewater release. The analysis reveals three distinct, coherent discursive ecosystems, each employing specific linguistic and rhetorical strategies—including naming, framing, modality, metaphor, and actor representation—to construct divergent realities of the same event. These discourses are not neutral reportage but are deeply embedded within and reflective of their respective national political contexts, institutional roles, and perceived strategic imperatives.

4.1 Japanese Outlets

In the landscape of Japanese media, the discourse surrounding the Fukushima discharge functions less as mere reporting and more as a sophisticated mechanism for risk mitigation. This is achieved by intertwining a meticulously curated technical lexicon with the perceived "finality" of international institutional backing. Crucial to this discursive strategy is the recurrent deployment of specific linguistic anchors.

A hallmark of Japanese reporting is the unwavering adherence to high-certainty modals that frame the discharge as a mundane industrial procedure. Note how NHK (24/8/2023) underscored that the operation commenced only “after safety was confirmed” (安全性が確認された上で), pointedly asserting that “tritium concentrations remain below standard limits” (トリチウム濃度は基準値を下回る). This is not merely descriptive; it is a Normalization. By anchoring the narrative in quantified metrics—numbers and thresholds—the media leans on scientific positivism to dampen visceral public apprehension. The calculated preference for "Treated Water" (処理水) effectively severs the link between the effluent and its radioactive pedigree, reclassifying what was once "waste" into a strictly regulated byproduct.

Within this linguistic landscape, the IAEA transcends the role of a mere consultant. It is cast instead as the provider of a moral and technical "finality" that shields national policy

from scrutiny. Yomiuri Shimbun (1/9/2023) highlights the government's pledge of "ensuring transparency based on scientific evidence" (科学的根拠に基づき透明性を確保) while leaning heavily on the "IAEA seal of approval" (IAEA のお墨付き). This manifests as a strategic use of Intertextuality. By positioning the IAEA as the "Final Arbiter," Japanese media constructs a narrative where the discharge is no longer a unilateral state decision but a globally sanctioned mandate. Such phrasing effectively shifts the conversation from the ethics of the act to the technicalities of monitoring, thereby neutralizing domestic and international dissent.

To safeguard the national image, Japanese media employs a set of evaluative adjectives that systematically categorize foreign criticism as irrational or biased. Sankei Shimbun (30/8/2023) dismissed China's countermeasures as "unjust measures" (不当な措置) that "ignore science" (科学を無視) and are driven by "politically motivated thoughts" (政治的な思惑). Such phrasing exemplifies a De-legitimization Strategy. By tethering "science" exclusively to Japan and "political emotion" to China, the media constructs a stark binary opposition. This linguistic maneuver reframes the international conflict not as a dispute over environmental safety, but as a clash between scientific enlightenment (Japan) and unscientific hostility (China).

4.2 Chinese Outlets

China's discourse is interwoven with geopolitical narratives and historical distrust of Japan (Sun et al., 2024). This is manifested through official denunciations of Japan's unilateral decision, portrayal of the release as an ecological danger, and countermeasures like seafood import bans—a stance echoed in domestic public opinion (Xu et al., 2023). Terminology is strategically employed to frame Japan as irresponsible and China as a protector of regional ecological security (Chen & Pan, 2025). Media discourse is alarmist and moralizing, using lexical, tonal, and representational patterns to emphasize danger, contamination, and ethical violation.

Central to this approach is a naming as political contestation strategy. Chinese media

categorically reject the Japanese nomenclature, systematically employing “核污染水” (nuclear-contaminated water) or “核污水” (nuclear wastewater), as seen in Xinhua’s report on the “正式启动福岛第一核电站核污染水排海” ("official commencement of Fukushima Daiichi nuclear-contaminated water sea discharge") (24/8/2023). This is not a passive choice of synonym but an active discursive refusal. The label functions to re-lexicalize the substance, permanently tethering it to its origin in catastrophe ("nuclear") and asserting an inherent, unprocessed state of danger ("contaminated/polluted"). This directly contests the Japanese narrative of scientific management, reframing the issue from one of waste processing to one of ongoing pollution. The charge of conceptual manipulation is made explicit in outlets like the National Business Daily, which accuses Japan of “偷换概念” ("stealing and changing concepts") by using the term “处理水” (treated water) (24/8/2023).

This lexical foundation is fortified by a moral lexicon as diplomatic weapon. The tone is saturated with evaluative language drawn from official diplomatic rhetoric, transforming policy criticism into moral indictment. Phrases such as “极端自私，极不负责任” ("extremely selfish and extremely irresponsible") and “一意孤行” ("obstinately sticking to one's course") are recurrent, as cited in responses from China’s Foreign Ministry (China News Report, 24/8/2023). This language performs a dual function: it attributes negative agency and character to the Japanese government while simultaneously positioning China as a guardian of international norms and collective welfare. The intertextual weaving of official condemnations—such as “中方对此表示坚决反对和强烈谴责” ("China expresses resolute opposition and strong condemnation")—into news reports merges media narrative with state policy, presenting a monolithic front of opposition.

The discourse further employs catastrophic metaphor for emotional mobilization. To translate technical risk into visceral public concern, media deploy potent imagery. The Pacific Ocean is framed not as a disposal site but as a shared commons violated, encapsulated in the forceful metaphor: “太平洋是世界的太平洋，不是日本的下水道” ("The Pacific Ocean belongs to the world, it is not Japan’s sewer") (People’s Daily Online, 13/4/2022). Temporal metaphors like branding the release date a potential “人类灾难日” ("disaster day for humanity")

(China News Report, 24/8/2023) serve to historicize and dramatize the event, elevating it from a procedural story to an epochal threat. Such metaphors bypass rational calculation, appealing directly to a sense of shared vulnerability and injustice.

In its representation of actors, Chinese coverage establishes a rigid dichotomy of responsibility. Japanese entities are consistently portrayed as agentive but duplicitous; for instance, the government is depicted as one that “无视国际社会的强烈质疑和反对，单方面强行启动” ("ignored the strong doubts and opposition of the international community and unilaterally forced the start") (National Business Daily, 24/8/2023). In stark contrast, Chinese authorities, experts, and affected groups are represented as reactive, victimized, or authoritative truth-tellers. References to “清华模拟结果” ("Tsinghua University simulation results") (Guangdong Local News, 23/8/2023) provide a countervailing scientific authority to challenge the IAEA, constructing a domestic epistemic foundation for the risk narrative. This polarized actor representation funnels all culpability toward Japan while legitimizing China’s stance as both victim and responsible stakeholder.

Finally, the discourse utilizes a modality of certainty to foreclose debate. Predictions of impact are articulated with high-certainty modals, presenting contested futures as inevitable facts. Reports state the discharge “将会影响整个太平洋乃至全球海域” ("will affect the entire Pacific and even global waters") and “必将长期受到国际社会谴责” ("will inevitably face long-term condemnation from the international community") (China National Radio, 23/8/2023; National Business Daily, 24/8/2023). This grammatical strategy of using “将会” (will) and “必将” (will inevitably) eliminates hedging, rhetorically converting potential scenarios into predetermined outcomes. It serves to amplify the perceived immutability and severity of the consequences, leaving no discursive space for the safe or normalized outcome posited by Japanese sources.

In summary, Chinese media craft a tightly coherent discourse that operates on moral, emotional, and geopolitical registers. Through the strategic integration of contested naming, condemnatory tone, vivid metaphor, dichotomous actor representation, and definitive modality,

it systematically deconstructs the technocratic framing advanced by Japan. The resulting narrative constructs the discharge as an act of irresponsible pollution, a moral failing, and a transboundary threat, thereby firmly situating China in the role of a principled challenger and protector within the regional and global order.

4.3 Western Outlets

The most striking textual pattern in the Western media sample is the lack of fixed names. Media outlets alternated between "treated" and "radioactive," describing the leak as manageable but still posing a risk. Reuters referred to the water as "treated water," describing the treatment process as "filtered + diluted + released" (Reuters, 22/08/2023). The Guardian also used the phrase "treated radioactive water" concluding with descriptions of the leak's scale, such as "500 Olympic-sized swimming pools," which reads technical explanation rather than sensational alarm (The Guardian, 15/02/2023). Meanwhile, some articles retained strong historical or moral overtones. For example, the Financial Times called the leak "legacy of the 2011 disaster" (Financial Times, 24/08/2023), while The Washington Post took a sharper approach, describing it as "decades of deliberate radioactive pollution of the marine environment" (Washington Post, 22/08/2023).

These naming choices are closely linked to the tone, metaphors, and modality of the reporting. A substantial proportion of the reports employ descriptive language, relying on data, infrastructure, and a step-by-step explanation. For example, the AP News reported the leak as "release … via undersea tunnel," while NPR similarly described it as "through an underground tunnel" (AP News, 03/11/2023; NPR, 22/08/2023). In this reporting style, metaphors is often used to visualize the scale (e.g., "Olympic-sized swimming pools") and predict the disaster (The Guardian, 15/02/2023). Furthermore, the reporting modality tends to be more subtle. Some headlines are presented as questions, such as "How risky is…?" (Financial Times, 24/08/2023) and "Is… safe?" (Scientific American, 23/06/2023), indicating that the assessment is open rather than final.

Furthermore, actor representation is shaped by who is quoted and how authority is staged. Tokyo Electric Power Company (TEPCO) and the Japanese government are typically portrayed as managers of a long-term process ("over decades, long-term decommissioning"), while the International Atomic Energy Agency (IAEA) is repeatedly positioned as the oversight body, with its assessment summaries showing "went well" and "no issues were observed" (PBS NewsHour, 23/08/2023; PBS NewsHour, 23/10/2023). Meanwhile, fishermen, environmental organizations, and neighboring countries are frequently mentioned, expressing concerns and anxieties, while Western discourse takes a middle ground: a science-oriented explanation, but public concerns are still reflected in the text.

When compared with Japanese and Chinese coverage, Western reporting occupies a middle ground, informed by science yet attentive to public sentiment, structured by technical frames yet open to debate. Through varied terminology, balanced actor portrayals, cautious modality, and the inclusion of multiple perspectives, Western media construct the wastewater discharge as a technically intricate, socially contested, and internationally significant environmental issue. This approach reflects the pluralistic norms that frequently shape journalism in Western contexts.

4.4 Cross-Regional Comparison of Media Discourse

The cross-regional analysis reveals that media in Japan, China, and Western contexts construct fundamentally different narratives of the same event through distinct discursive strategies. Japanese media employ a strategy of technocratic normalization. By consistently using the term "treated water" and prominently featuring the IAEA as a final scientific authority, the discourse is engineered to depoliticize the discharge, frame it as a routine and safe technical procedure, and shield national policy from ethical challenge. Criticism, particularly from China, is systematically delegitimized as unscientific or politically motivated.

In stark contrast, Chinese media engage in geopolitical and moral condemnation. The adamant use of "nuclear-contaminated water" permanently ties the effluent to the 2011

catastrophe and frames the discharge as an act of irresponsible pollution. This lexical choice is reinforced by intertextual reliance on domestic scientific sources and official diplomatic rhetoric, merging media narrative with state policy. The discourse constructs a rigid dichotomy: Japan is portrayed as a duplicitous and irresponsible agent, while China is positioned as a victimized guardian of regional ecological security and international norms.

Western media coverage, meanwhile, reflects a mode of pluralistic deliberation. It avoids monolithic framing by employing varied terminology—oscillating between “treated” and “radioactive”—which acknowledges both the scientific process and persistent public concern. Sourcing is deliberately pluralistic, incorporating voices from the IAEA, TEPCO, independent scientists, environmental NGOs, and affected communities. This approach constructs the event not as a settled fact but as a complex, socially contested socio-technical dilemma, embodying a journalistic norm of balanced debate amid uncertainty.

Ultimately, these three discursive ecosystems do not neutrally report an event but actively manufacture different realities: a managed industrial procedure, an act of transboundary ecological aggression, and an open environmental controversy. This divergence underscores the powerful role of media discourse in shaping international risk perception, assigning political responsibility, and reinforcing deep-seated geopolitical alignments.

V. Conclusion

This study has employed Critical Discourse Analysis to systematically investigate how news media from Japan, China, and Western nations construct divergent realities of the Fukushima wastewater discharge. The analysis reveals that the starkly different terminologies—from Japan's "treated water" to China's "nuclear-contaminated water" to the West's varied usage—are not merely semantic preferences but core elements of broader, ideologically driven narratives.

The findings demonstrate three coherent and internally consistent discursive systems. Japanese media engage in technocratic normalization, leveraging scientific authority and institutional intertextuality to frame the discharge as a safe, routine, and internationally sanctioned procedure, thereby mitigating perceived risk and delegitimizing opposition. Chinese media deploy a strategy of geopolitical and moral condemnation, using alarmist lexicalization, vivid metaphor, and dichotomous actor representation to construct the event as an act of irresponsible ecological aggression, thereby positioning China as a principled defender of global norms. Western media, in contrast, exhibit a pattern of pluralistic deliberation, balancing technical descriptions with public concern through heterogeneous terminology, multivocal sourcing, and cautious modality, presenting the issue as a complex and socially contested environmental dilemma.

These discursive constructions underscore a central tenet of CDA: language is a form of social practice that shapes and is shaped by power relations. The media narratives are deeply embedded within their respective national political contexts and strategic imperatives. They do not passively report an event but actively manufacture distinct public understandings, influencing international risk perception, assigning blame, and reinforcing geopolitical alignments.

This research contributes to the field by providing a targeted, comparative analysis of the interplay between lexicalization, intertextuality, and actor representation—a gap identified in

existing scholarship. It moves beyond cataloguing framing differences to explain how these differences are produced through specific, interrelated linguistic strategies. The study highlights the critical role of media discourse in contemporary environmental governance, where scientific uncertainty intersects with high political stakes. It suggests that in transnational disputes, the struggle over naming and framing is not a peripheral debate but a fundamental battleground for legitimacy, authority, and public consent.

Future research could expand on this foundation by incorporating a broader temporal scope to trace discursive shifts, integrating audience reception studies to examine how these media frames are interpreted by the public, or applying a similar comparative CDA framework to other global environmental controversies. Ultimately, this analysis affirms that in the geopolitics of risk, the power to name is a power to define reality itself.

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VI. Appendices:

Appendix A: Coded News Articles

Japan Media:

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Appendix B: Coding Table for 33 News Articles

CDA Dimension			Textual features				
Region	Outlet	Date	Naming	Tone	Actor Representation	Metaphors/Imagery	Modality
Japan	Yomiuri Shimbun 読売新聞	24/8/2023	“Treated water” “処理水”	Neutral	Gov’t/TEPCO: responsible managers	“Scientific basis.” “科学的根拠”	High certainty
	NHK news	5/9/2023	“Treated water” “処理水”	Objective reporting	Domestic Fishery Groups: concerned parties.	“Safety standards.” “安全基準”	High certainty
	Asahi Shimbun 朝日新聞	25/8/2023	“Treated water” “処理水”	Supportive	Neighboring Countries: Source of diplomatic	“Restoration of trust.” “信頼回復”	Medium certainty
	Mainichi Shimbun 毎日新聞	10/9/2023	“ALPS-treated water” “ALPS 処理水”	Calm response	Experts: assurers-providing scientific backing	“Safety assurance.” “安全確保”	High certainty
	Sankei Shimbun 産経新聞	30/8/2023	“Treated water” “処理水”	Diplomatically firm	Chinese Government: Irrational opponent	“Unjust measures’ “不当な措置”	Medium certainty
	NHK World (English)	26/8/2023	“Treated water” “処理水”	Internationalized	International Community: accepting and understanding party	“Scientific review.”	High certainty
	Jiji Press 時事通信	15/9/2023	“ALPS-processed water” “多核種除去設備の処理水”	Process emphasis	TEPCO: implementer-focus on strict procedural adherence	“Strict standards.” “厳格な基準”	High certainty

	Yomiuri Shimbun 読売新聞	28/9/2023	“Treated water” “处理水”	Economic outlook	Fisherman Industry: affected parties	“Reputational damage countermeasures.” “風評被害対策’	Medium certainty
	Asahi Shimbun 朝日新聞	5/10/2023	“Treated water” “处理水”	Call for rational dialogue	Non-Government Organizations: Skeptics-their views mentioned but framed by Gov’t arguments	“Diplomatic efforts.” “冷静な議”	Medium certainty
	Sankei Shimbun 産経新聞	20/9/2023	“Treated water” “处理水”	National pride	Japanese Fisherman: resilient victims	“Food safety.” “食の安全”	High certainty
	Jiji Press 時事通信	1/10/2023	“ALPS-treated water” “ALPS 处理水”	Procedural details	Third-Party Independent Agencies: watchdogs-assuring neutrality	“Thrid-party verification.” “第三者検証”	High certainty
China	People's Daily Online 人民网	13/04/2022	“Nuclear-contaminated water” “核污染水”	Critical	Japan's government: careless China & Korea gov: protective	“The Pacific Ocean is not a dumping ground for Japan.” “太平洋不是日本的下水道”	High certainty
	Xinhua net 新华网	24/08/2023	“Radioactive wastewater” “核废水”	Alarm	Japan's government and TEPCO: irresponsible Japan’s citizens: victims	“This is an issue that concerns our very survival.” “关乎生存的问题”	High certainty
	China National Radio 央广网	23/08/2023	“Nuclear-contaminated water” “核污染水”	Critical	Japan’s government: irresponsible Tsinghua researchers：authoritative	“Areas with higher concentrations are shown in red” “高浓度区域呈红色”	High certainty
	China.com.cn 中国网	24/08/2023	“Radioactive wastewater” “核废水”	Alarm	Japan’s government： careless Chinese authorities： protective Scientists China and Germany：authoritative	“A day of human catastrophe” “人类灾难日”	High certainty

	News Channel of Hong Wang (Red net) in Hunan Province 湖南红网新闻频道	22/08/2023	“Radioactive wastewater” “核废水”	Critical	Japan’s government and TEPCO: careless Global community: victims	“There is no untouched land”, “a food chain that passes contamination upward at every level.” “没有净土”, “逐级上传的食物链”	High certainty
	National Business Daily 每日经济新闻	24/08/2023	“Nuclear-contaminated water” “核污染水”	Morally condemning	Japan’s government and TEPCO: irresponsible Industries: impacted	“The seawater shows two distinct colors.” “海水呈现两种颜色”	High certainty
	China National Radio 央广网	24/08/2023	“Nuclear-contaminated water” “核污染水”	Warning	China Experts: authoritative Japan’s government: irresponsible	“The spread that extends across the Pacific Ocean.” “跨越太平洋的扩散”	High certainty
	Zhejiang News 浙江新闻	25/08/2023	“Radiation risk” “放射性物质”	Concerned	Japan’s government and TEPCO: irresponsible Consumers: anxious	“Devastating entire industries.” “毁掉产业”	High certainty
	China Daily Online 中国日报网	07/07/2023	“Radioactive wastewater” “核废水”	Critical	IAEA： procedural and technical, but insufficiently Japan’s government： untrusted Experts: transparency challengers	“It is impossible to support this wholeheartedly.” “无法举双手赞成”	Medium–High certainty + uncertainty of impact
	National Business Daily 每日经济新闻	24/08/2023	“Nuclear-contaminated water” “核污染水”	Warning	Japan’s government： deceptive Chinese authorities： protective Korea Protesters： justified	“The nuclear-contaminated water flows into the Pacific Ocean within just seventeen minutes.”	High certainty
Western	PBS NewsHour	23/08/2023	‘treated’ water	Balanced	TEPCO And Japan’s government：management local fishermen / fisheries	“Over decades, long-term decommissioning”	low certainty

					industry：criticial		
	Reuters	22/08/2023	“Radioactive water”	Neutral	Japan’s government and TEPCO: cautions Fishery organization and China: critical	“Treated water” + “filtered + diluted + released”	High certainty
	The Guardian	15/02/2023	“treated radioactive water”	Scientific	Japan’s government And IAEA: authoritative Neighboring countries: worried Fishery organization: anxious	“500 Olympic-sized swimming pools”	Low certainty
	AP News	03/11/2023	“Radioactive water”	Neutral	TEPCO And Japan’s government: decisive China And South Korea: objective	“Release … via undersea tunnel”	High certainty
	Washington Post	22/08/2023	“Radioactive water”	Neutral	TEPCO And IAEA: authoritative Environment protection organization: worried China And South Korea: objective	“Decades of deliberate radioactive pollution of the marine environment”	High certainty
	Financial Times	24/08/2023	“Radioactive water”	Cautious	TEPCO And Japan’s government and IAEA: Leading Fisherman And public: anxious	“legacy of 2011 disaster”	Low certainty
	BBC	30/07/2025 3	‘treated’ water	Balanced	IAEA And Japan’s government:authoritative Public And Local community:worried	Advanced Liquid Processing System（ALPS）	High certainty
	NPR	22/08/2023	‘treated’ water	balanced	Japan’s government：decisiveFisherman And Neighboring countries:worried	release through an underground tunnel	Low certainty
	Scientific	23/06/2023	‘treated’ wastewater	Scientific	TEPCO: Managing	“More than 1.3 million cubic	Low

	American				Scientists: critical International community and experts: having different opinions	meters of seawater … contaminated with 64 radioactive elements”	certainty
	PBS NewsHour	23/10/2023	“treated radioactive wastewater”	Neutral	IAEA: authoritative Neighboring countries and fishers:objective	“went well / no issues were observed”	Low certainty
	TIME	06/02/2023	“Wastewater”	Emotional	TEPCO And Japan’s government：decisiveLocals / angry locals：objectiveExperts:skeptical	“The people of the Pacific are people of the ocean”	Low certainty
	The Economist	13/07/2023	‘treated’ water	Appraisive	Japan’s government：decisive China, South Korea and Pacific island countries: Protest	“Rowing about”	Low certainty